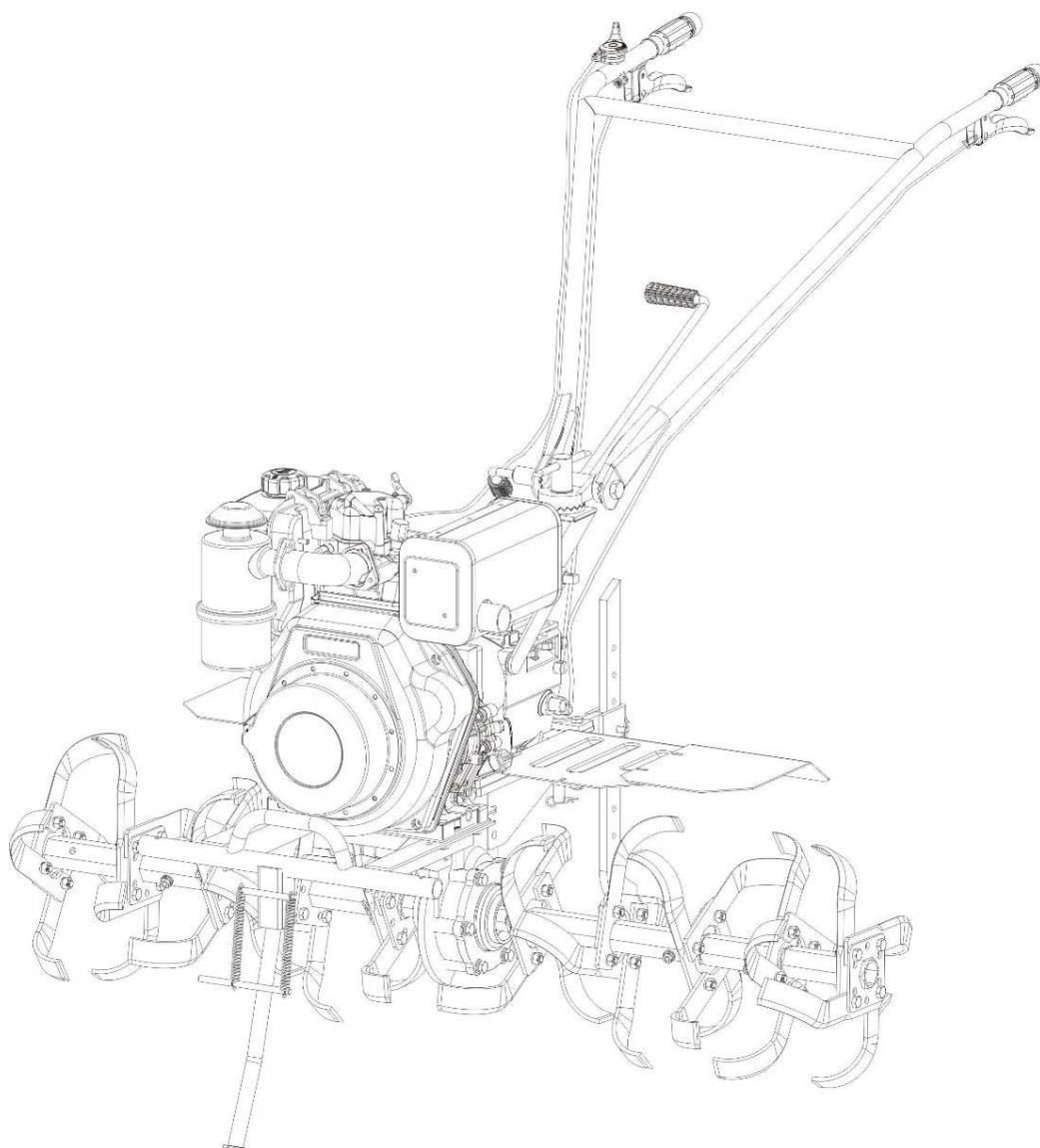


MULTIFUNCTIONAL TILLER MACHINE

OWNER'S MANUAL



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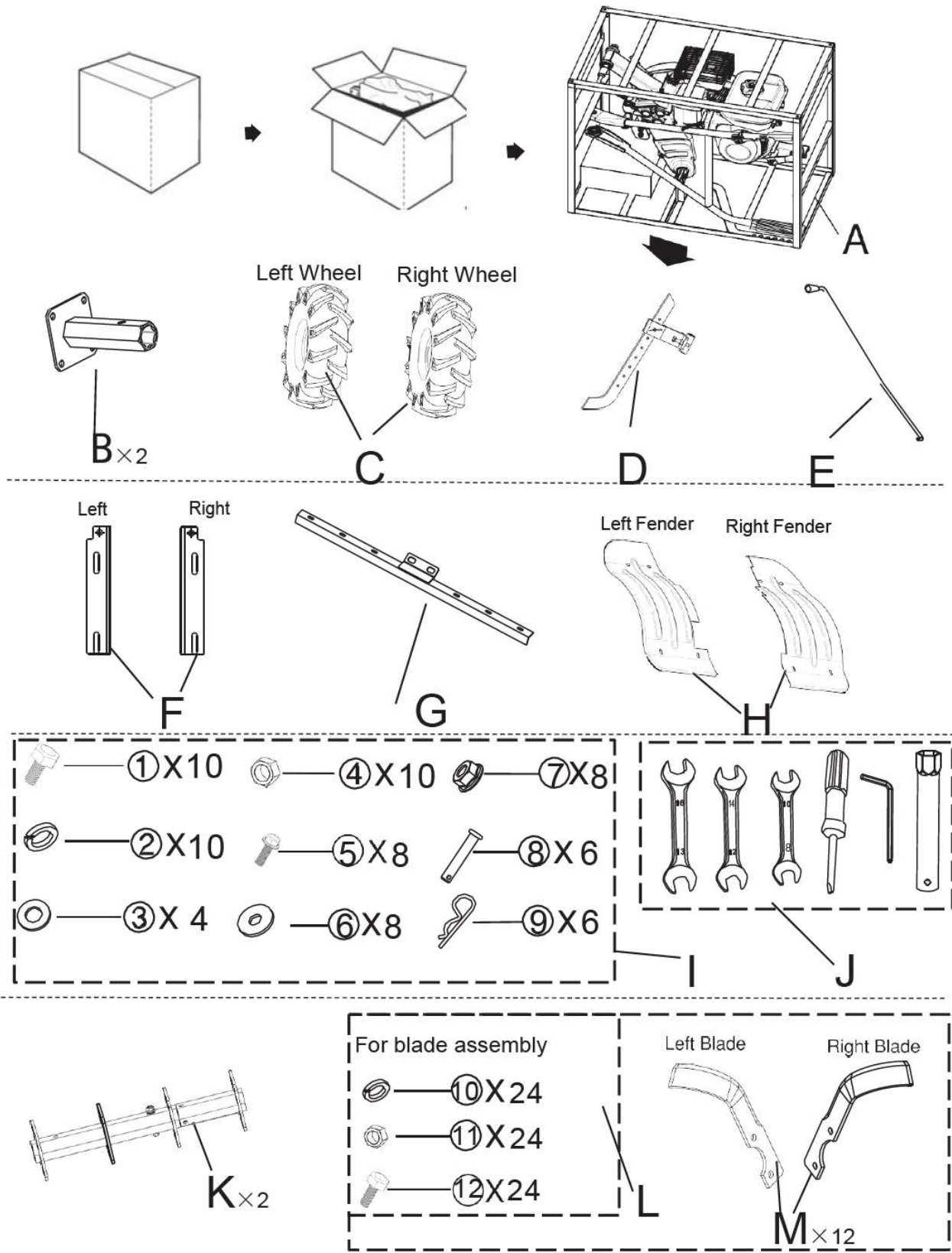
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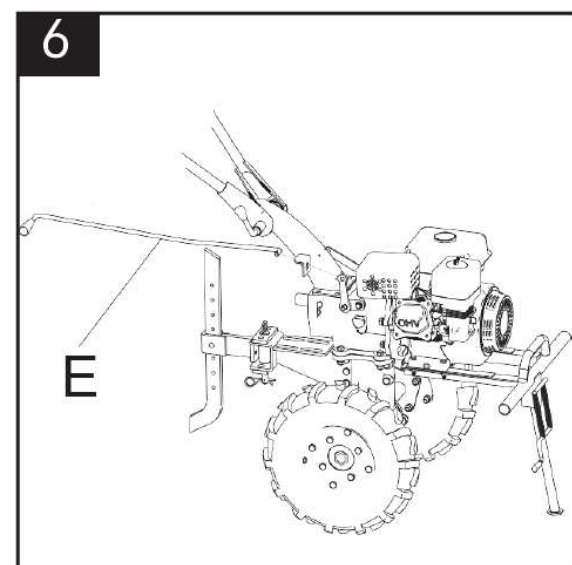
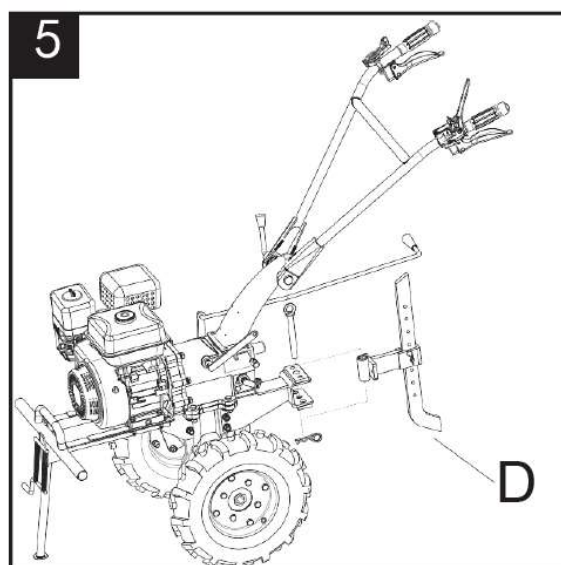
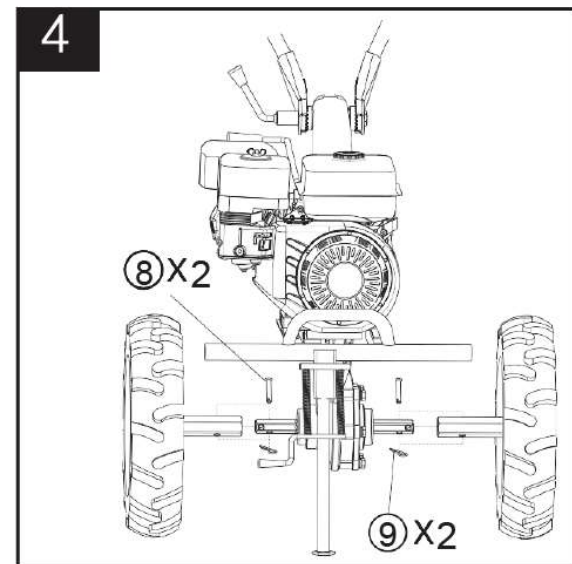
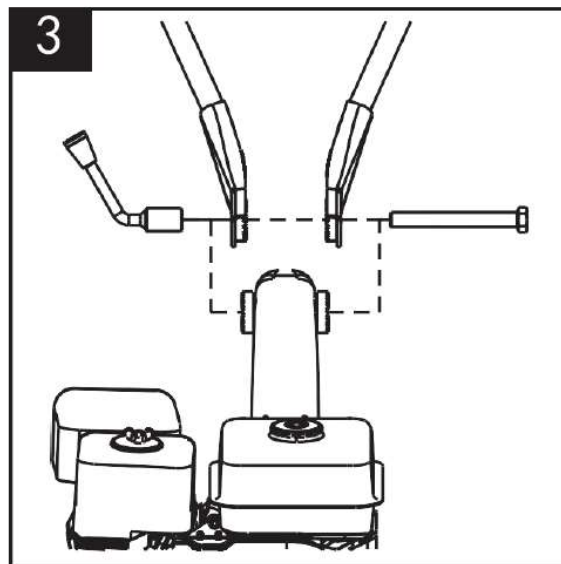
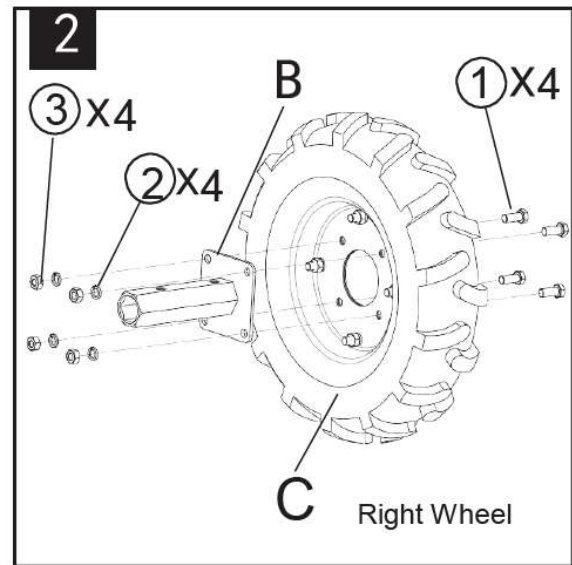
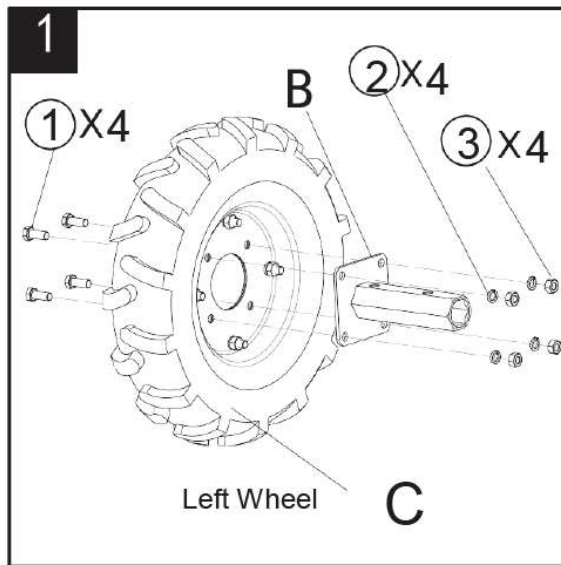
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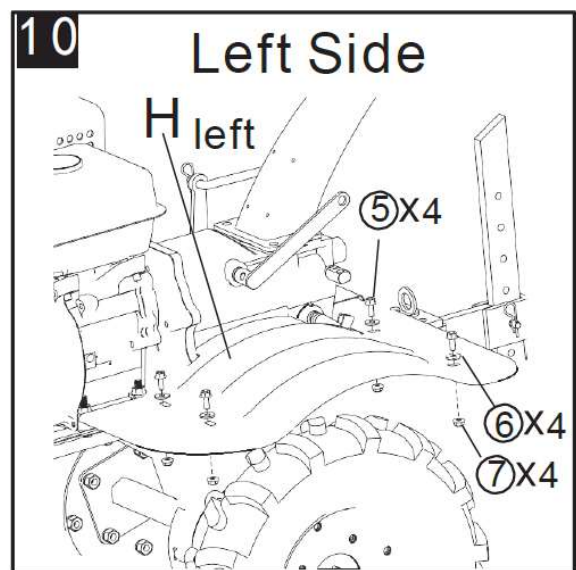
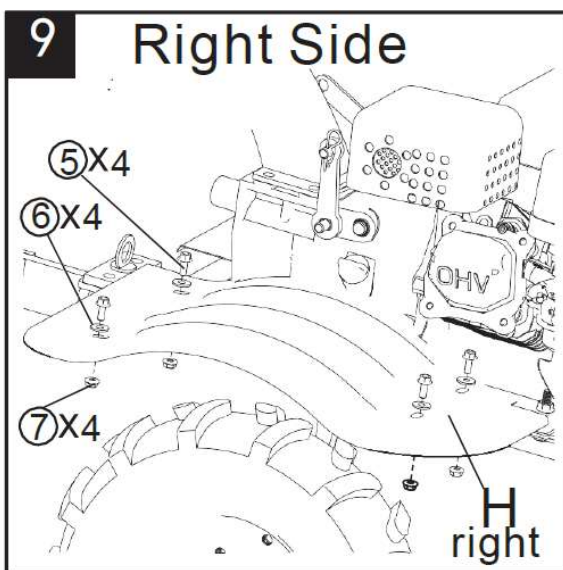
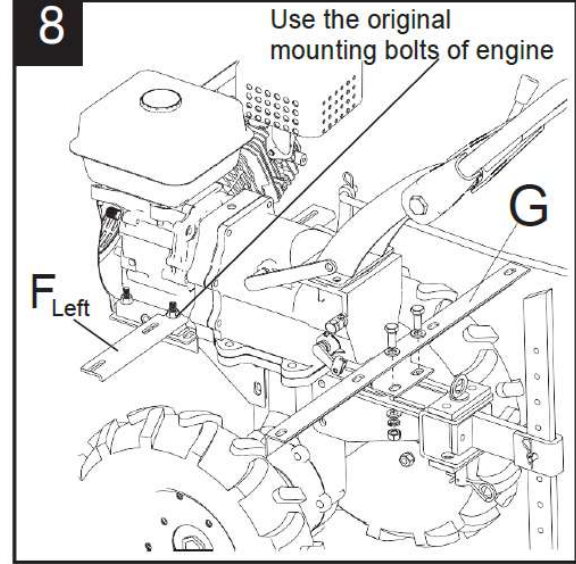
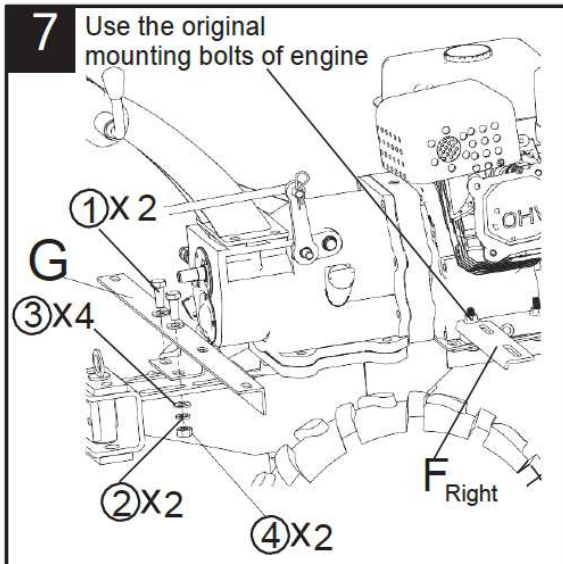
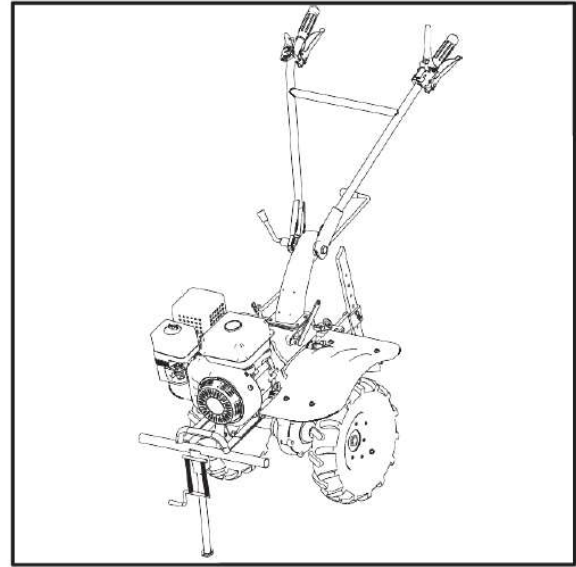
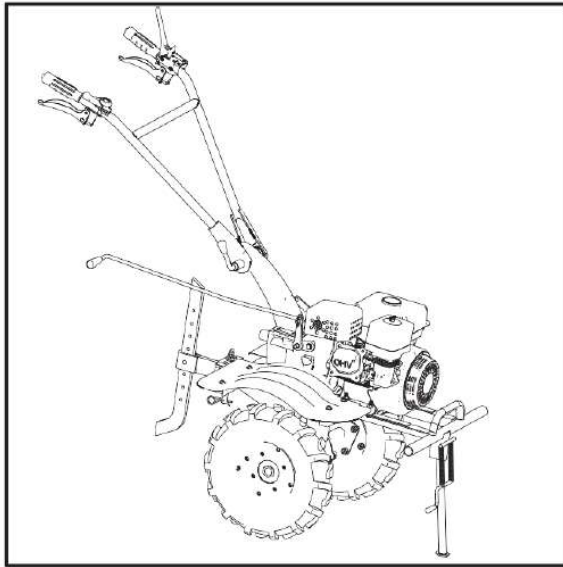
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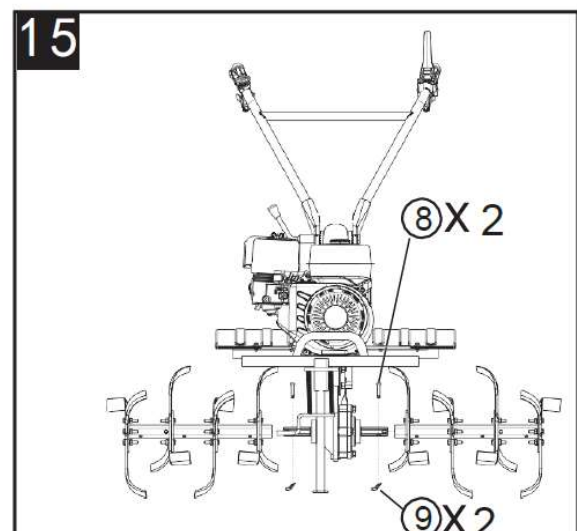
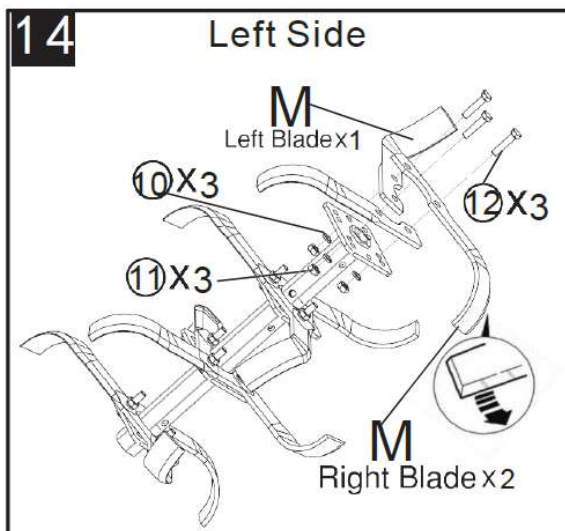
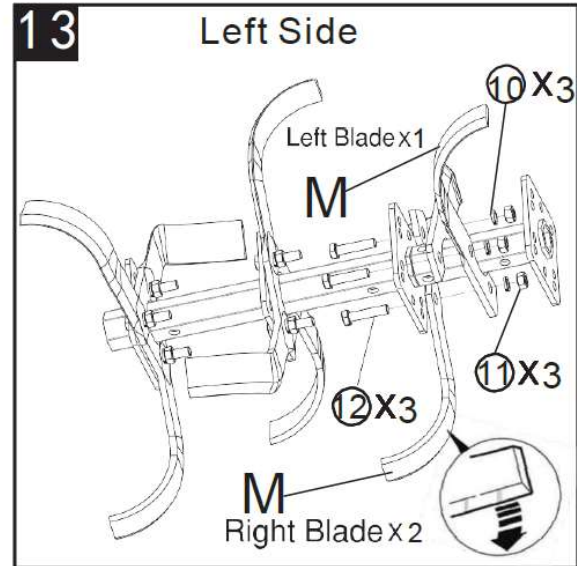
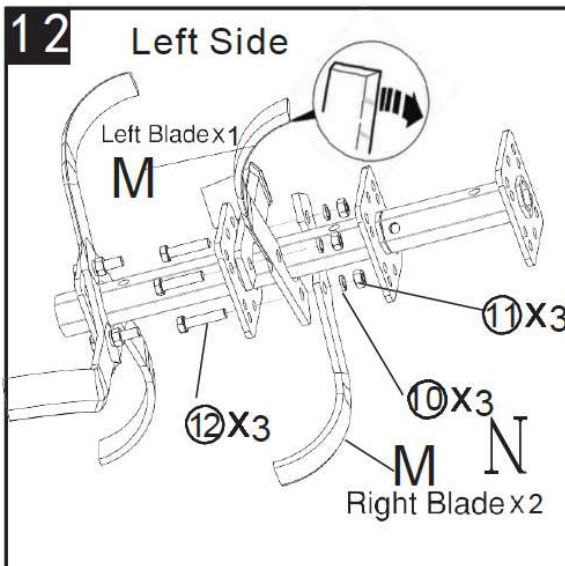
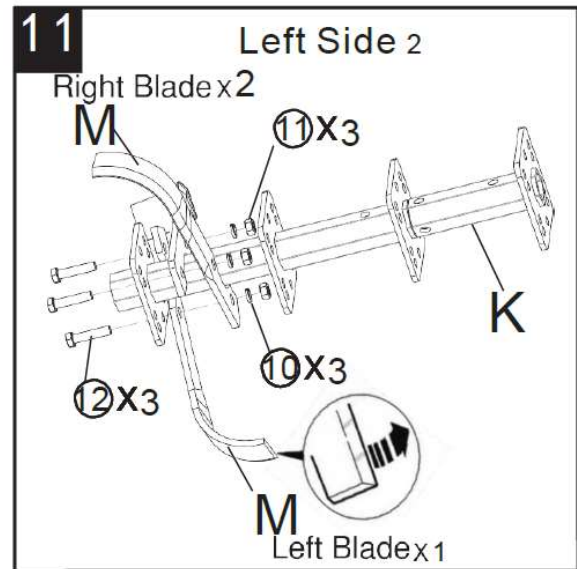
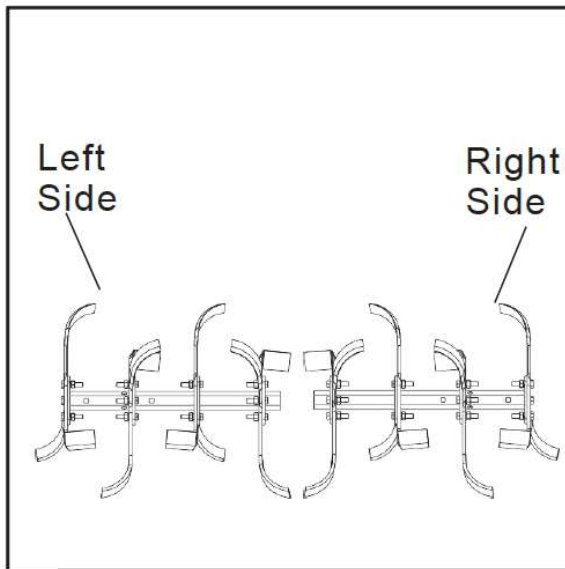
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1. Model 210P/270P



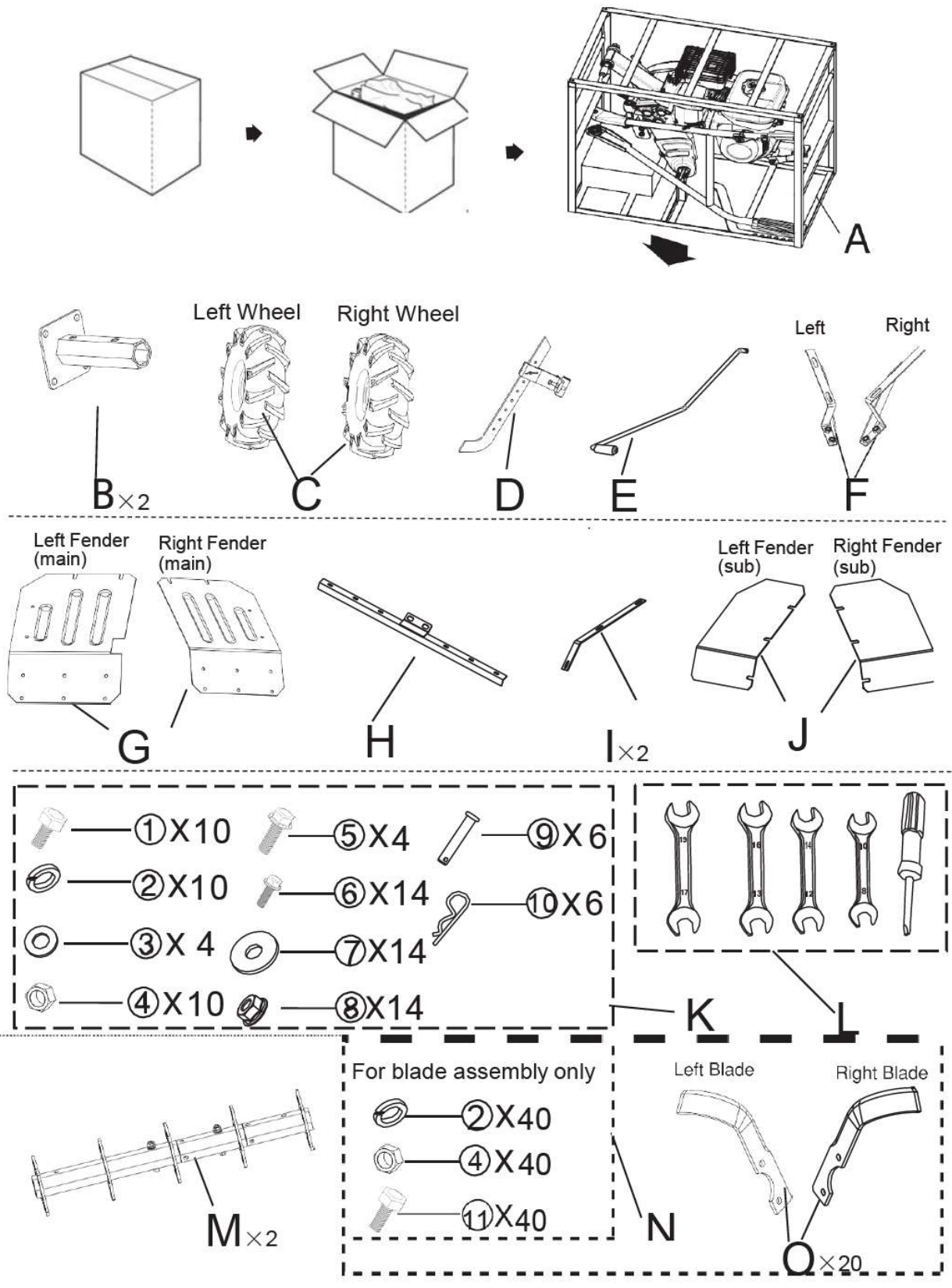


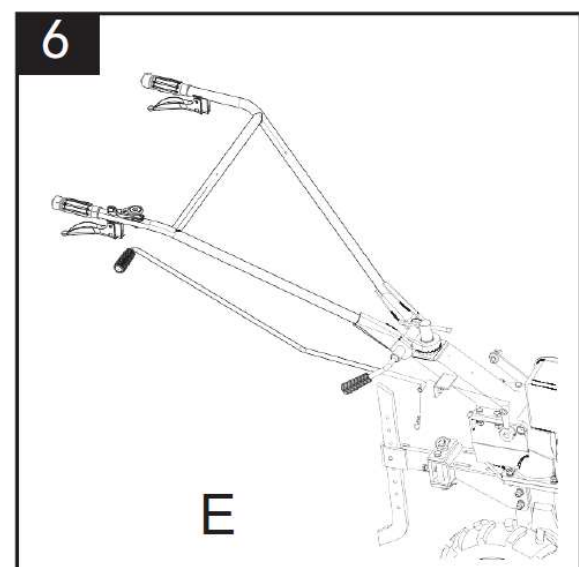
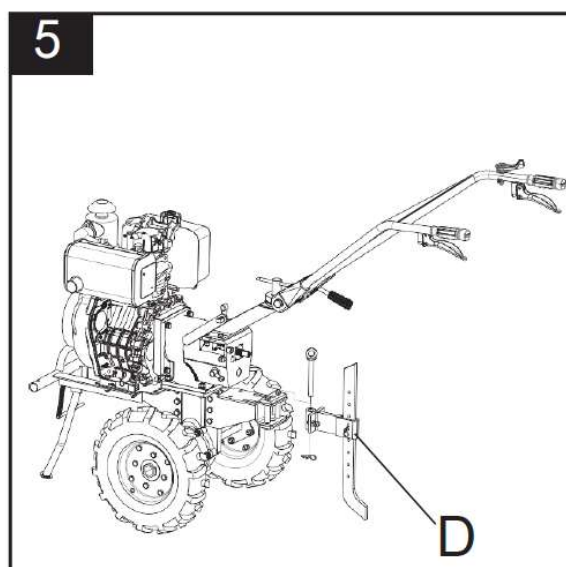
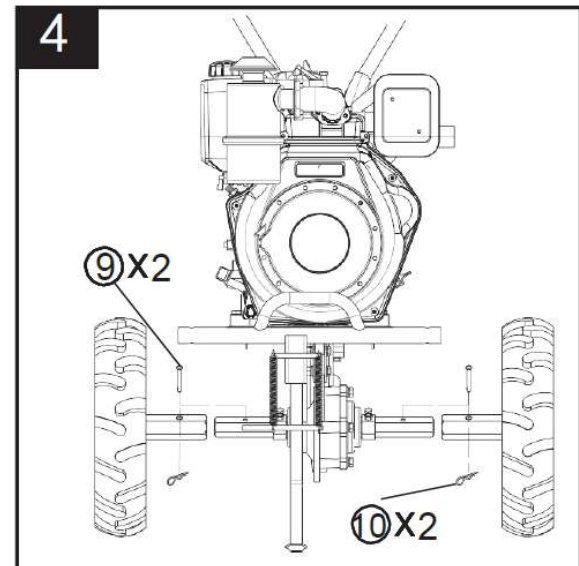
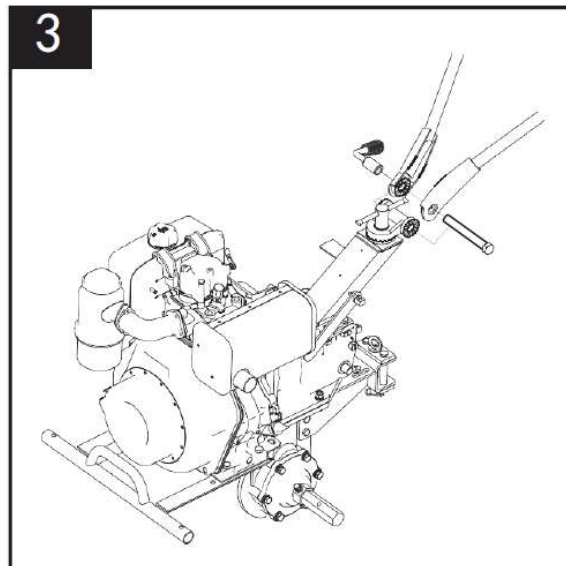
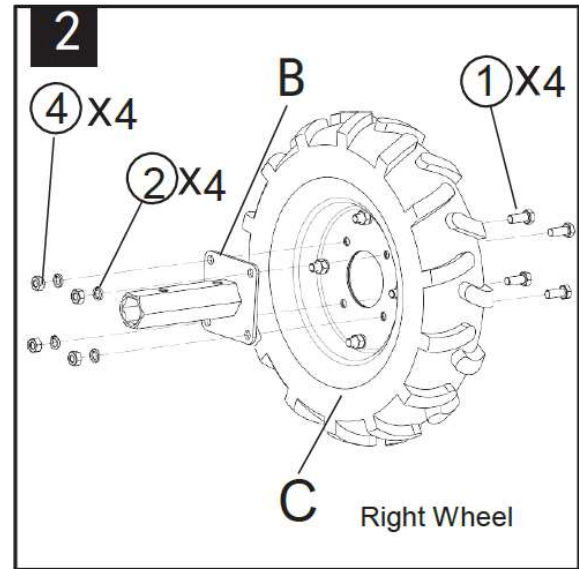
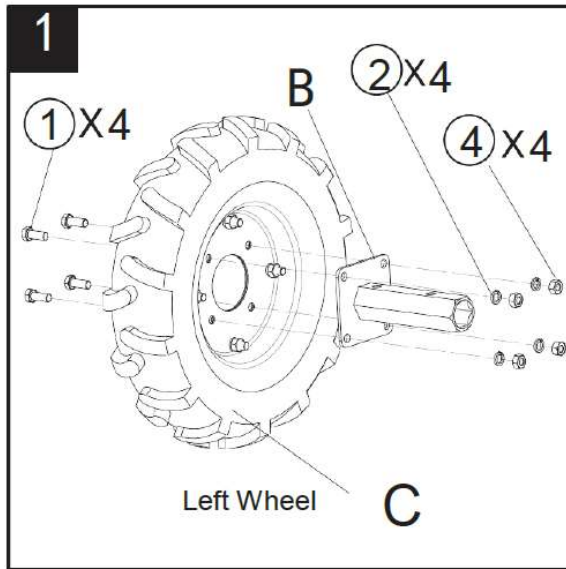


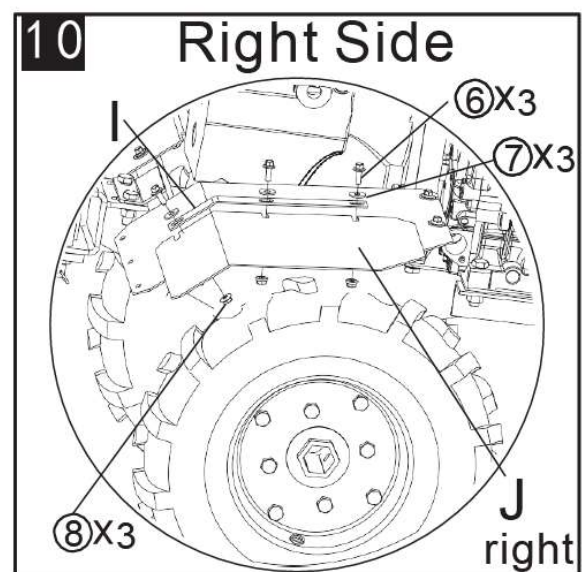
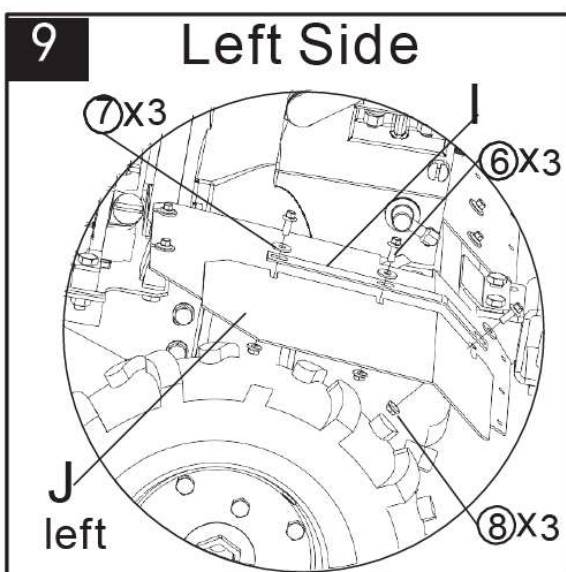
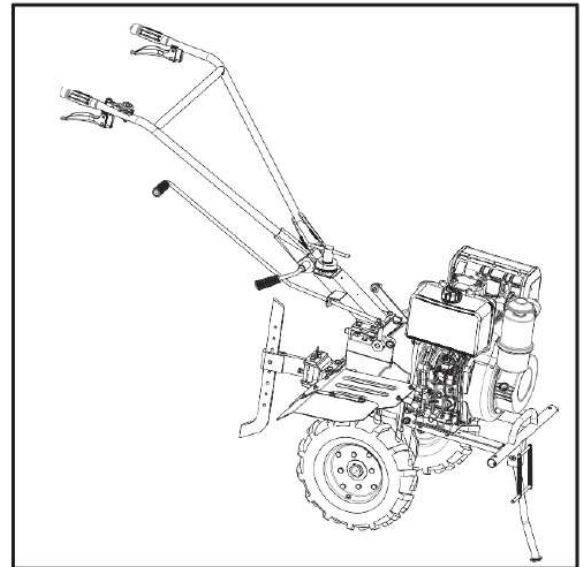
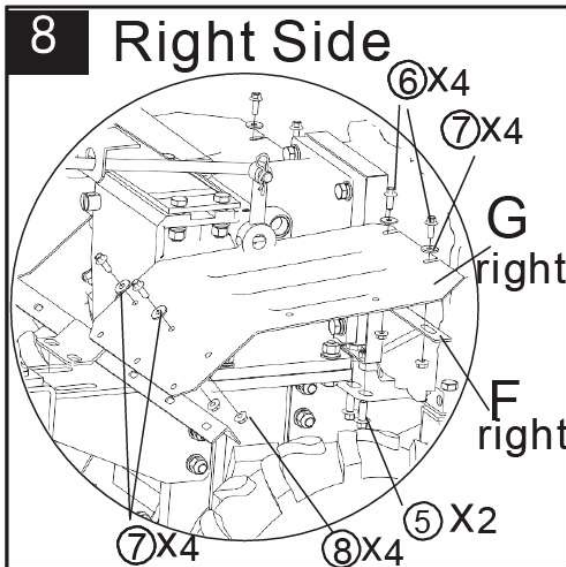
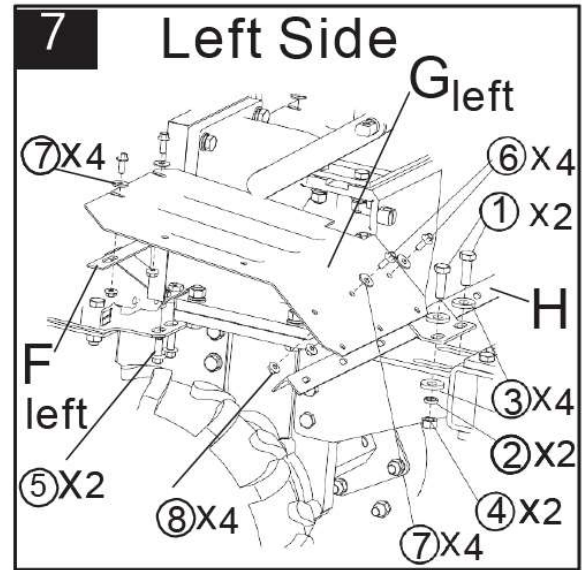
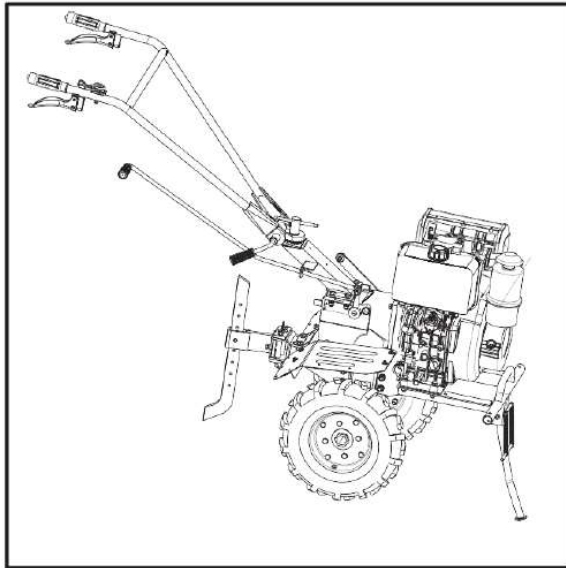


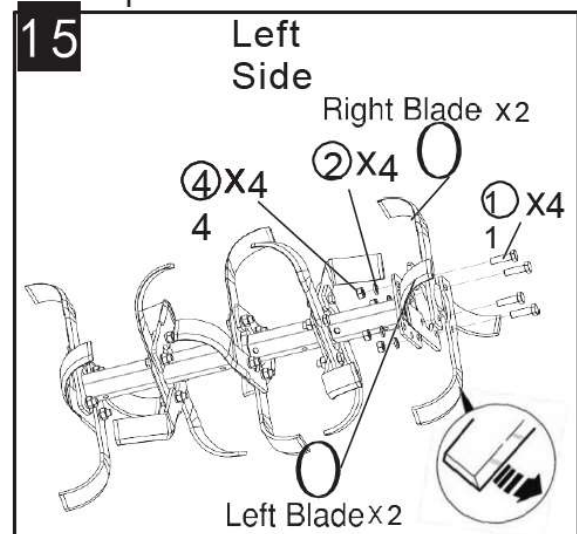
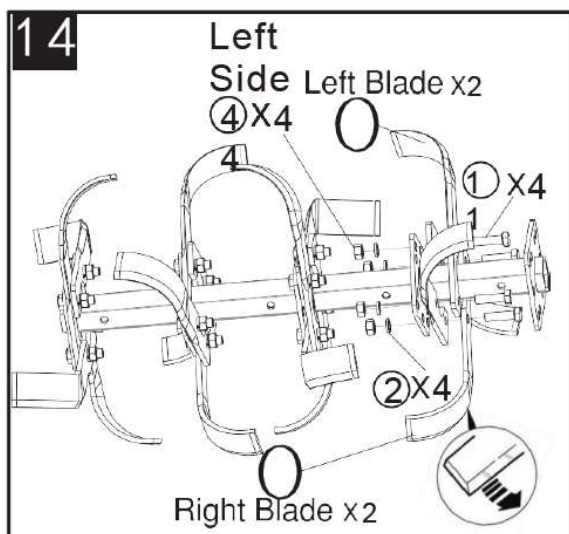
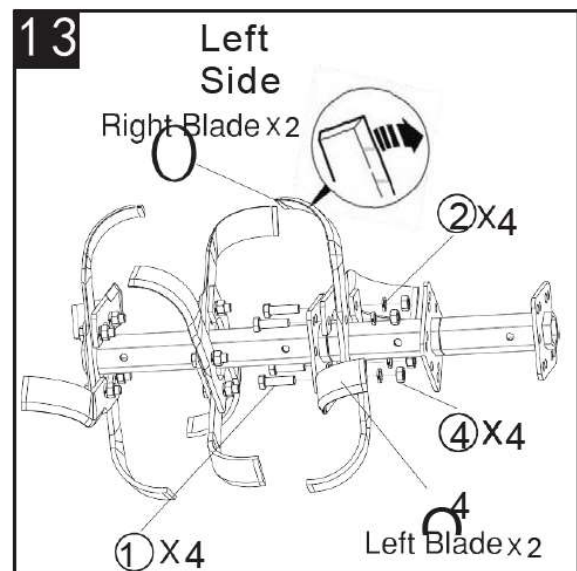
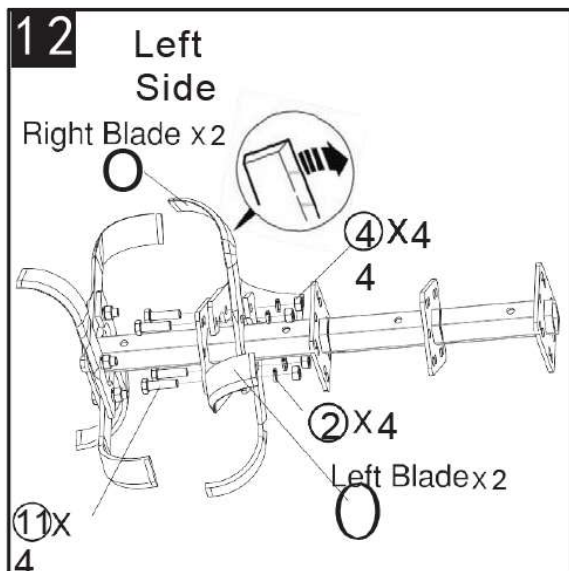
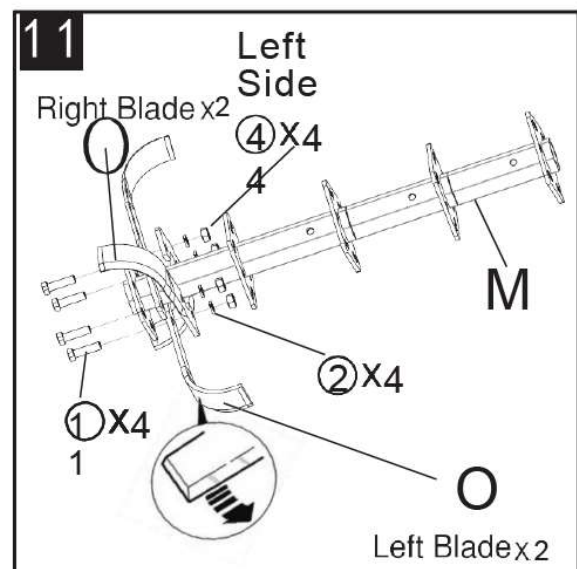
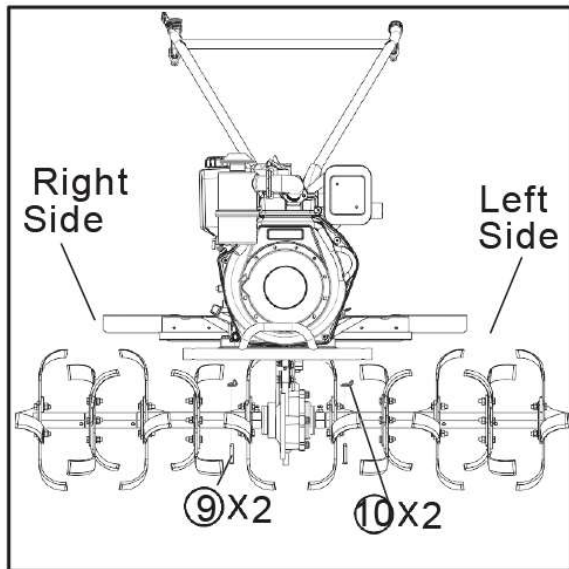
Notes: 1. Blade edge must forward.
2. Left and right blades are symmetric.

2. Model 250D/310D/420DE









Notes: 1. Blade edge must forward.
2. Left and right blades are symmetric.

1. Notice for safety

- 1.1 Read the manual carefully before use the machine, follow the manual to do running-in, adjustment and maintenance.
- 1.2 Before using the machine:
 - 1.2.1 Check if there is engine crankcase or gearbox leakage; check the lubricant oil level and oil quality of crankcase and gearbox, replenish or change clean oil in time.
 - 1.2.2 Please refuel the engine by clean diesel. Please refuel when the engine is stopped in ventilated place. Be careful not to let the fuel contact high temperature surface, electric parts or rotation parts. Don't overfill the tank to avoid spilling, please check if the fuel spilled or leaked, wipe and dry the spilled fuel before start the engine. Fasten the fuel tank cap after refueling, no fire or smoking at refueling place, fuel storage place or working site to prevent fire hazard.
 - 1.2.3 Check if each fastener is fastened; if each moving parts is loosened, interfered or blocked; if the rotating direction is right.
 - 1.2.4 Check all exposed rotating parts, moving parts if there is reliable protection, if the safety signs are complete.
 - 1.2.5 Check the clutch, blades, and cables if they are deformed or excessively worn.
 - 1.2.6 If there is abnormal phenomenon, please eliminate it and do test run. There should not be interference, abnormal sound, excessive vibration during the test run. Engine speed should be within the limit of 3600r/min, over speed is not allowed. When change the parts involved in safety, please follow the manual or under the guide of professional technician.
- 1.3 Only adults who is trained and know the usage rules of the tiller can operate it.
- 1.4 The operator is not allowed to work after drinking, in spite of illness or overtired.
- 1.5 The operator should fasten the cloth and cuff while working, who is with long hair should also wear protective hat.
- 1.6 It is prohibited to modify the parts of tiller which is involved in safety and operation. The operator should concentrate while working.
- 1.7 After safety check, you may start the tiller, the tiller cannot bear heavy load when it just cold started, especially new tiller or after major repair.
- 1.8 Prevent the tiller with blades to run on the concrete road, stone path or gravel road. While tilling, avoid to bump into stone or other hard objects in order not to damage the blades.

- 1.9 Pay attention to working performance and sound of each part during operating. If any part connection is abnormal, there is any abnormal sound, must stop the engine immediately. When the engine is running, remedy the fault is forbidden.
- 1.10 Prevent the tiller from toppling over when working.
- 1.11 Reverse gear when your back is less than 2 meters from the field edge.
- 1.12 Please take off the resistance rod before switch to reverse gear.
- 1.13 Please be careful of gearbox and engine leakage during work, if it happens, stop the engine and check the tiller immediately, remove the fault in time to avoid environmental pollution.
- 1.14 Clear the weed and dirt on blades after stop the engine, it's prohibited to clear the blades by stick or hand when the engine is still running.
- 1.15 After working, please clean the dirt, weed or oil on the tiller.
- 1.16 Transport the tiller with rubber wheels between different fields.
- 1.17 Please check the tightness of the bolts on the blade assembly, bearing holder and other moving parts at regular intervals.

Tiller Safety Introductions

1. Training

- 1.1 Please read the user manual in depth, learn about control system and correct usage of the machine, master how to stop the tiller and rapidly hold the clutch.
- 1.2 Never allow children use the machine, never allow anyone use the machine before read the manual.
- 1.3 Keep other people away from the working area, especially children and animals.

2. Preparation

- 2.1 Completely check the field before working, remove all other objects, e.g. stone.
- 2.2 Before start the engine, hold the clutch and shift into neutral gear.
- 2.3 Do not operate the machine unless wearing proper clothing. Wearing non-slip shoe could improve the foothold stability.
- 2.4 Take care with the fuel, it's flammable.
 - 2.4.1 Use appropriate container to keep the fuel.
 - 2.4.2 Do not refuel the engine while it is working or still warm.
 - 2.4.3 Please refuel outdoor carefully.
 - 2.4.4 Before start the engine, screw up the fuel tank cap, wipe the spilled diesel.
 - 2.4.5 No fire.
- 2.5 When the tiller is running, do not make any adjustment.
- 2.6 Wear safety goggles when prepare, operate or repair the tiller.

3. Operation

- 3.1 Keep your hands and feet away from the rotating parts.
- 3.2 When transporting the tiller on road, sidewalk or street, be careful of control, pay attention to traffic and be aware to potential danger.
- 3.3 If bump against any object, stop the engine immediately, check the tiller if there is damage, and repair the tiller before start working again.
- 3.4 Always be careful underfoot, avoid slipping.
- 3.5 If the machine vibrate abnormally, stop the engine and check the reason, vibration often indicates malfunction.
- 3.6 Before leave the control position, before clear the winded stuff on the blades or before repairing, adjusting and examine, must stop the engine first.

- 3.7 When the machine isn't under control, take any possible precautionary measures: separate the clutch, lower down the accessory, shift into neutral, stop the engine, and take off the start key.
- 3.8 Before clear, repair or examine the machine, the engine should be stopped, and make sure all moving parts stopped.
- 3.9 Do not run the engine indoor, the exhaust is harmful.
- 3.10 Ensure all protection are in position, without proper protections such as bumper and fender do not work with the tiller.
- 3.11 Keep children and pets away.
- 3.12 Do not overload the machine by large depth and high speed.
- 3.13 Do not travel too fast on slippery road. Watch behind carefully when move backwards.
- 3.14 Don't allow bystander get close to the machine.
- 3.15 When the engine is working, the muffler will be extremely hot, approaching or touching will be scalded.
- 3.16 Only the accessories and implements from original factory can be used.
- 3.17 Do not use the tiller when it's dim or the field of view is poor.
- 3.18 Take care when working on hard ground, the blade could be stopped by some objects.
When it happens, release the handle and do not try to control the tiller.
- 3.19 Do not work on steep hills.
- 3.20 Prevent overturn when the tiller is moving on slope.

4. Repair and storage

- 4.1 Keep the tiller, attachments and implements in good conditions.
- 4.2 Once in a while, check if the engine fixing bolts and other bolts are fastened to ensure the device is under safe working conditions.
- 4.3 Store the device indoor, away from source of fire. Cool down the engine first before store the tiller.
- 4.4 While store the tiller for long time, this manual should be kept together as crucial material.
- 4.5 Do not repair the device at will, unless you are experienced and with proper tools and disassembling, assembling guide.

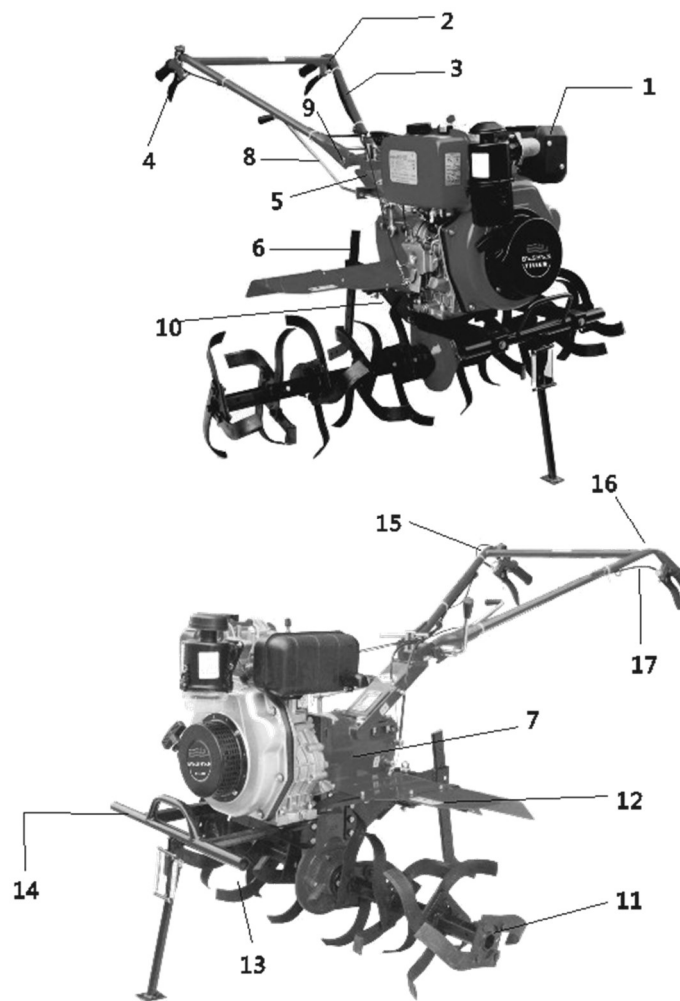
Survey on Tiller

1. Main Technical Parameter

Model			Unit	FWMT250D	FWMT310D	FWMT420DE	
Item	Parameter						
Engine	Model /Name			Diesel engine	Diesel engine	Diesel engine	
	Rated Power		KW	3.5	4.4	6.5	
	Max Power		KW	3.9	4.9	6.9	
	Rotating speed	Rate rotating speed			3600		
		Rotate speed of power outputshaft			3600		
	Displacement		ml	247	306	418	
	Ignition mode			Compression ignition			
	Start mode			Recoil starter			
	Fuel tank capacity		L	2.5	3.5	5.5	
	Engine oil capacity		L	0.75	1.1	1.65	
Tiller	Machine’s weight		Kg	≤95	≤105	≤120	
	Transmission mode			Full gear transmission			
	Connection mode			Direct Connection			
	Speed	Running speed	Fast	Km/h	11	11	11
			Slow		6.5	6.5	6.5
		Working Speed	Cultivating		0.5-1.3	0.5-1.3	0.5-1.3
					Reversing speed		2
	Blades	Max. radius of gyration		mm	180	180	180
		Total blades quantity		pcs	32	32	40
		Total groups of blades		pcs	8	8	10
		Blades for each group		pcs	4	4	4
		Diameter of the farming blades		cm	36	36	36
	Rotary blade type			Bent blade for dry land			
	Clutch mode			Wet multiple disc clutch land manual control			
	Tilling Depth		cm	≥15	≥15	≥15	
Tilling Width		cm	105	105	135		
Gear-box’s oil capacity		L	1. 5	2. 0	2. 0		

Parameter			Model	Unit	FWMT210P	FWMT270P	FWMT400P
Item							
Engine	Model /Name				Petrol engine	Petrol engine	Petrol engine
	Rated Power			KW	3.8	5.2	7.8
	Max Power			KW	4.4	5.5	8.3
	Rotating speed	Rate rotating speed			3600		
		Rotate speed of power outputshaft			3600		
	Displacement			ml	212	270	389
	Ignition mode				TDI/Spark Ignition		
	Start mode				Recoil Starter		
	Fuel tank capacity			L	3.6	5.0	6.5
	Engine oil capacity			L	0.6	1.0	1.1
Tiller	Machine’s weight			Kg	≤85	≤95	≤110
	Transmission mode				Full gear transmission		
	Connection mode				Direct Connection		
	Speed	Running speed	Fast	Km/h	11	11	11
			Slow		6.5	6.5	6.5
		Working Speed	Cultivating		0.5-1.3	0.5-1.3	0.5-1.3
					Reversing speed		2+1
	Blades	Max. radius of gyration		mm	180	180	180
		Total blades quantity		pcs	32	32	40
		Total groups of blades		pcs	8	8	10
		Blades for each group		pcs	4	4	4
		Diameter of the farming blades		cm	36	36	36
	Rotary blade type				Bent blade for dry land		
	Clutch mode				Wet multiple disc clutch land manual control		
	Tilling Depth			cm	≥15	≥15	≥15
	Tilling Width			cm	105	105	135
	Gear-box’s oil capacity			L	1. 3	1. 5	2. 0

2. General Structure



- | | |
|--------------------|--------------------------|
| 1. Engine | 10. Dragging Assembly |
| 2. Throttle | 11. Blade Assembly |
| 3. Handle | 12. Fender |
| 4. Clutch | 13. Engine Stand |
| 5. Handle Seat | 14. Bumper |
| 6. Resistance Road | 15. Throttle Cable |
| 7. Gear Box | 16. Emergency Stop Cable |
| 8. Shift Lever | 17. Clutch Cable |
| 9. Handle Shaft | |

Tilling

Assemble the tilling blades assembly on the gearbox output shaft for tilling work, pay attention to the difference of left and right side, fix the blades assembly by 8×50 pin.

Blade type	4 group		5 group	
	3 pcs	4 pcs	3 pcs	4 pcs
Blade quantity	24 pcs	32 pcs	30 pcs	40 pcs
Tilling width	1050 mm		1350 mm	
Applicable field	hard soil		wet soil after rain	dry land or with weed

1. First time assembling

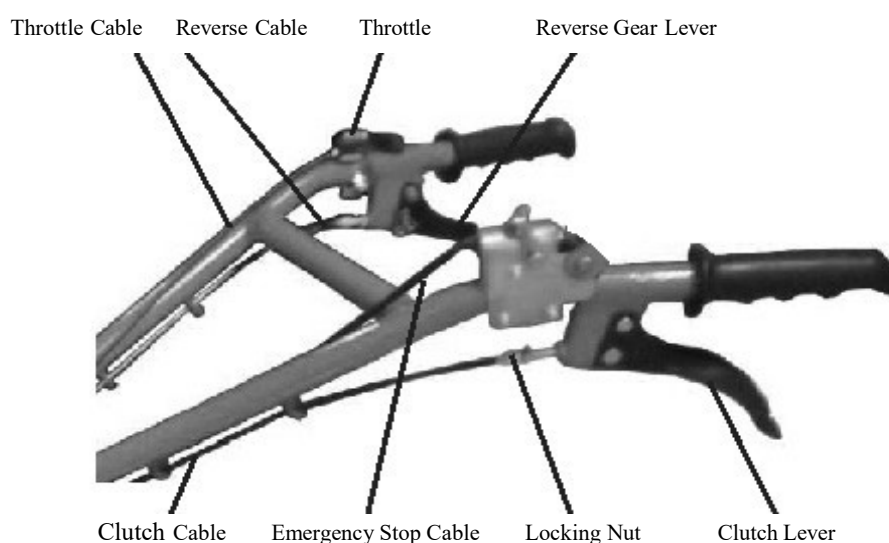
- 1.1 Assemble the wheel: put the wheel on both sides of the gearbox, use pin 8×50, pin clip 2×9×38 to fix them on the output shaft.
- 1.2 Dragging assembly: put the connection rod on the dragging assembly, put the fixing pin through the dragging assembly and connection rod, and fix with pin clip. Put the resistance rod into slot of the connection rod, use pin 8×50 and pin clip to fix.
- 1.3 Handle: match the fluted plate on the handle and handle seat, put M16×150 bolt through the handle from left to right, adjust the height of handle, tighten the locking lever.
- 1.4 Shift lever: put the shift lever through the supporting hole on the handle and the hole on the shift gear sheath of the gearbox, fix with pin clip, and shift into neutral.

2. Examination and refuel

2.1 Examination

2.1.1 Clutch cable adjusting.

- 2.1.1.1 Loosen the locking nut on the cable.
- 2.1.1.2 Screw the threaded rod clockwise until the rod is shortest.
- 2.1.1.3 Put the end of the cable into the cable fixture and make sure the cable is in the hole.
- 2.1.1.4 Press the clutch lever and tighten the locking nut.
- 2.1.1.5 Unscrew the threaded rod and press the clutch lever repeatedly, until clutch lever can return to normal position by the cable tension.



2.1.2 Reverse gear cable adjusting.

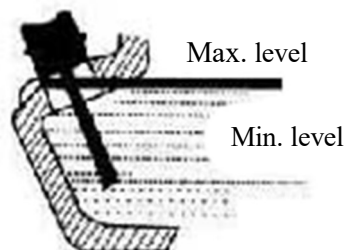
- 2.1.2.1 Loosen the locking nut on the cable.
- 2.1.2.2 Screw the threaded rod clockwise until the rod is shortest.
- 2.1.2.3 Check the control system, each lever (clutch, throttle, reverse gear) movement should be flexible, correct the improper situation.
- 2.1.2.4 Put the shift lever to neutral position.

2.2 Add lube oil.

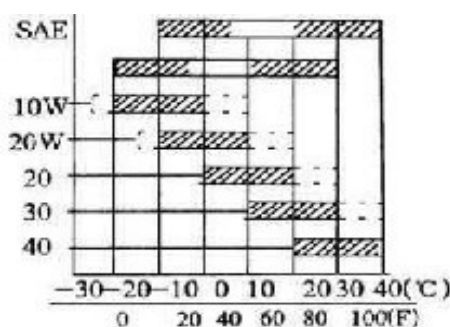
2.2.1 Choose lube oil according to environment temperature.

2.2.2 Add SAE10W-40 lube oil in the engine.

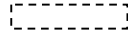
Place the tiller horizontally, add oil through the hole on top of the gearbox. When check the oil level, do not screw the dipstick.



volume	L/gal
	1.1(0.24)



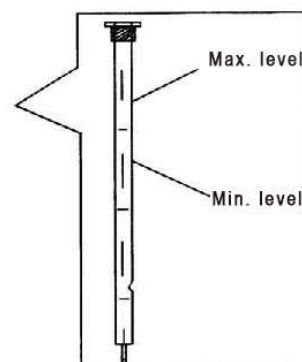
 Suggested

 Maximum

Diesel engine oil should be CC or CD grade based on A.P.I grades

2.2.3 Add 20# lube oil in the gearbox. Place the tiller horizontally, add oil through the hole on top of the gearbox. When check the oil level, do not screw the dipstick, oil level is between the max level and min level.

2.2.4 Add oil in the air filter, take off the bottom of air filter, add 0.1L 20# oil.



- 2.3 Refueling. According to engine type, fill the tank with 0# diesel or 92# gasoline.
- 2.4 Make preparations before starting the engine.

3. Starting

WARNING: Shift lever should be at neutral position.

- 3.1 Hold the red emergency stop lever, press down the locking catch to fix the lever.
- 3.2 Follow the engine manual procedure to start the engine.
- 3.3 The engine should idle for 2~3 minutes (1500-2000r/min).
- 3.4 Check if the engine is running normally.

4. Operation

ATTENTION: Must do running-in before working, running-in method refers to chapter 8.

4.1 Shift into slow gear.

- 4.1.1 Left hand hold the clutch lever to separate the clutch from the engine.
- 4.1.2 Right hand pull the shift lever back into slow gear, check if the gear shifting is in right position, then right hand hold the handle. Attention: don't hold the reverse gear lever.
- 4.1.3 Slowly loosen the clutch lever to engage the clutch with the engine, the tiller will run in slow speed.
- 4.1.4 Right hand increase the throttle properly, the tiller will run in slow speed at 4.5km/h.

4.2 Shift into fast gear.

- 4.2.1 Left hand hold the clutch lever to separate the clutch from the engine.
- 4.2.2 Right hand push the shift lever forward into fast gear, check if the gear shifting is in right position, then right hand hold the handle. Attention: don't hold the reverse gear lever
- 4.2.3 Slowly loosen the clutch lever to engage the clutch with the engine, the tiller will run in fast speed.
- 4.2.4 Right hand increase the throttle properly, the tiller will run in fast speed at 8km/h.

4.3 Shift into reverse gear.

- 4.3.1 Left hand hold the clutch lever to separate the clutch from the engine.
- 4.3.2 Right hand put the shift lever to the middle into neutral gear, check if the gear shifting is in right position.
- 4.3.3 Then the right hand hold the reverse gear lever, loosen left hand, the tiller will run backwards.
- 4.3.4 After moved back to the ideal position, left hand hold the clutch lever first, the tiller will stop, then right hand loosen reverse gear lever.

4.4 Shifting during working.

First reduce the throttle but don't let the engine stop, then hold the clutch lever to separate it, shift after the tiller stopped working.

5. Turning.

Turn the handle to let the tiller turn direction. Do not hold the reverse gear lever by mistake in order to avoid damage the gearbox.

6. Stop

- 6.1 Hold the clutch lever to separate the clutch from the engine.
- 6.2 Shift into neutral first, loosen the clutch lever, reduce the throttle to minimum to stop the machine.
- 6.3 When you need to stop the engine, please follow the manual of diesel engine. Attention: Stop the tiller on flat ground.

7. Emergency stop

- 7.1 During operating, while the throttle is at minimum position, if the clutch is still not separated (because of certain fault), the machine cannot stop, emergency stop is needed at this moment.
- 7.2 Press the red color emergency stop lever and release it at once, the lever will unlock itself, the engine will stop.
- 7.3 If the engine cannot stop by other method, press the decompression lever, the engine will flameout and stop working.

Maintenance

During working, because operation, friction wearing and load change, loose bolts, worn parts, engine power reduction and fuel consumption increase could happen on the tiller. In order to reduce the above incident, we have to do maintenance work regularly and strictly, to keep the tiller in good function, extend its working life.

1. Running-in

1. Engine running-in please follow the guide of the engine manual.
2. New tiller or after overhaul, should work without any load for 1 hour and work with light load for 5 hours, then drain the lube oil from engine crankcase immediately. Refill oil again and running-in for another 4 hours (refer to chapter 3), the tiller will be ready to work.

2. Technical maintenance

1. Maintenance before and after work
 - 1.1 Listen and observe each part of the tiller, discover abnormal sound, overheat or loose bolt.
 - 1.2 Check if there is leakage in the engine or gearbox.
 - 1.3 Check the oil level of engine and gearbox, if it is between the maximum and minimum marks.
 - 1.4 Clear the soil, weed and oil on the tiller and accessories in time.
 - 1.5 Make working notes.
2. Grade 1 maintenance - every 150 working hours
 - 2.1 Do all maintenance work above.
 - 2.2 Clean the gearbox case and replace oil.
 - 2.3 Check and adjust clutch, gear shifting and reverse gear system.
3. Grade 2 maintenance - every 800 working hours
 - 3.1 Do all maintenance work for every 150 working hours of Grade 1.
 - 3.2 Check all the gears and bearings, replace the worn ones.
 - 3.3 Other parts of the tiller like the blades or joint bolts, replace damaged ones.
4. Technical maintenance - every 1500-2000 working hours
 1. Go to the after service center or the dealer's to disassemble the tiller, clear and check the tiller, replace worn parts or repair appropriately.
 2. Let the technician check the clutch core, especially friction plate.
5. Engine maintenance please follow the guide of the engine manual.

3. Maintenance Time Table

The interval regards to the working hours or the used months, depends on which is reached first.	Daily	8 hours under half load	1 month or 20 hours	3months or 150 hours	12 months or 1000 hours
Check and fasten the bolts and nuts	●				
Check oil level and fill it	●				
Clear and replace the oil		● 1st time	● 2nd time	●	
Check oil leakage	●				
Clear the soil, weed and oil	●				
Remove malfunction	●				
Adjust control parts	●				
Clutch Friction Plate					●
Gear and bearings					●

4. Tiller storage

To prevent rust, follow the steps below to store the tiller for long time.

1. Follow the engine manual to store the engine.
2. Clear the dust and dirt on the surface.
3. Drain the oil from gearbox and fill it with new oil.
4. Apply anti-rust paint to surface which is without painting and not aluminum.
5. Store the product indoor at ventilated, dry and safe place.
6. Keep the attachment tools, certificate of qualification and user's manual properly.

Adjustment

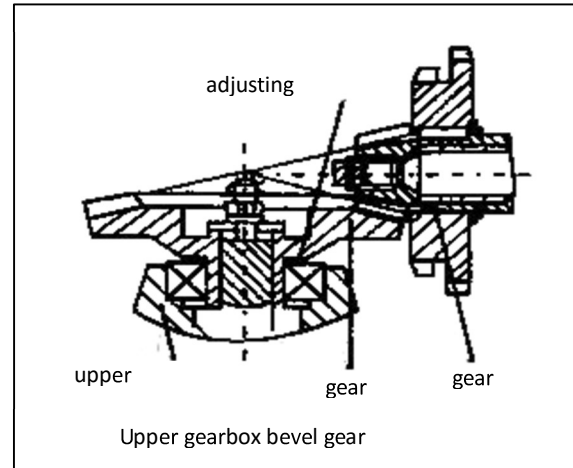
1. Bevel gear pair adjusting

When the bevel gear pair transmit power abnormally or make too much abnormal sound, please check and adjust, follow the method below:

1. Upper gearbox bevel gear pair clearance adjusting.

1.1 When the bevel gear pair side clearance $< 0.05\text{mm}$, add metal gasket between upper gearbox and lower gearbox to increase clearance.

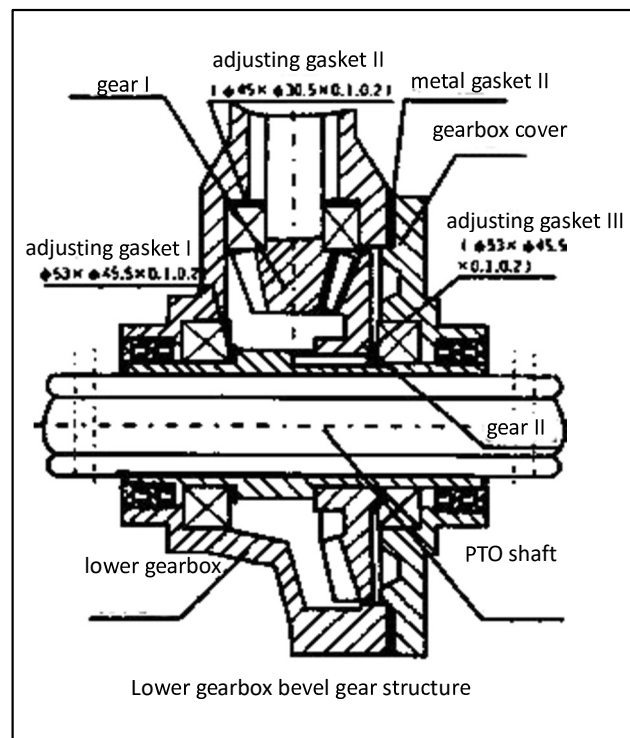
1.2 When the bevel gear pair side clearance $> 0.3\text{mm}$, adjust clearance between bearing and gear II to $0.05 \sim 0.10\text{mm}$.



2. Lower gearbox bevel gear pair clearance adjusting.

2.1 When the bevel gear pair side clearance $< 0.05\text{mm}$, add adjusting gasket I $0.2 \sim 0.3\text{mm}$ to increase clearance. And change metal gasket II and adjusting gasket III to make gear II axial clearance to $0.05 \sim 0.15\text{mm}$.

2.2 When the bevel gear pair side clearance $> 0.3\text{mm}$, reduce adjusting gasket I, and make gear II axial clearance to $0.05 \sim 0.15\text{mm}$; or add adjust gasket II, and make gear II axial clearance to $0.05 \sim 0.15\text{mm}$.



2. Reverse gear and cable adjusting

When the tiller move backwards abnormally, then adjust the reverse gear lever and cable, refer to chapter 3.

Attention: 1. Hold and release the handle lever 2-3 times, check the gear engagement situation, adjust until it is qualified.

2. During the tiller is running, release reverse gear lever, reverse gear should return immediately, there should not be abnormal sound of gear crashing.

3. Clutch cable adjusting

After using for certain time, the clutch performs abnormally because the friction plates and clutch fork wear, then adjust the clutch cable referring to chapter 3.

Attention:

1. Hold and release the clutch lever 2-3 times, check the clutch engagement situation, adjust until it is qualified.
2. If adjusted repeatedly still did fix the problem, the clutch fork or friction plate might wear out, please change the friction plate or clutch fork.
3. To avoid damage the mechanism, disassembling the clutch is forbidden.

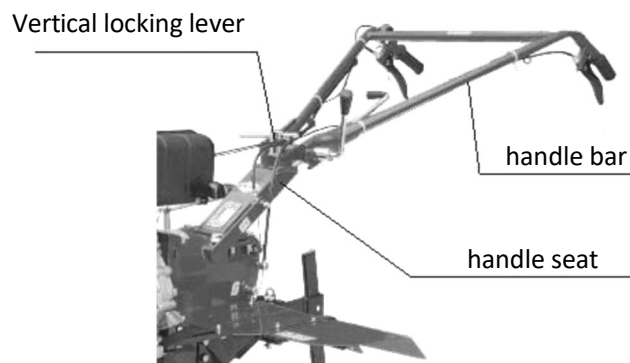
4. Throttle cable adjusting

When turning the throttle switch, the engine speed does not change effectively, then adjust the throttle cable referring to chapter 3.

- Attention:
1. Turn the throttle switch 2-3 times, check and confirm engine speed changes.
 2. The throttle cable and end of the thread should be fixed by bolt firmly.

5. Handle adjusting

Depending on you height and specific agricultural work requirements, the handle can be adjusted both horizontally and vertically.



1. Handle vertical adjusting.
 - 1.1 Loosen the locking lever, un-match the teeth between handle and handle seat.
 - 1.2 Choose a comfortable position of the handle as your habit.
 - 1.3 Tighten the locking lever again to match the handle and handle seat.
2. Handle horizontal adjusting.
 - 1.1 Loosen the locking lever on handle seat, un-match the flute plates of handle seat.
 - 1.2 Turn the handle left or right as your need.
 - 1.3 Tighten the locking lever again to match the flute plates.

Malfunctions and Solutions

1. Engine malfunction and solutions, please refer to engine manual.

2. Clutch malfunction and solutions.

Attention: Do not disassemble clutch assembly by your own.

Description	Reason	Solution
Clutch dysfunction	Clutch lever failure	Repair or replace
	Clutch cable failure	Replace
	Clutch fork not adjusted well	Adjust the cable or replace the fork
	Clutch fork, shaft not fixed well	Repair or replace
	Clutch fork break or bend	Replace
	Friction plate failure	Replace
	Spring failure	Replace
	Friction plates cannot touch the bearing surface	Add adjusting gasket under the bearing
	Bearing failure	Replace bearing and add lube oil in gearbox
Bad connection (holding the clutch lever, PTO rotates slowly)	Spring fatigue and failure	Replace
	Clutch fork shaft cannot rotate	Clear the shaft rotating surface
	Clutch cable not adjusted well	Adjust the clutch cable

3. Gearbox malfunction and solutions.

Description	Reason	Solution
Shifting problem	Bolt or nut behind the main shaft are loosen	Disassemble the bolt, nut and sheath behind the main shaft, put them back and screw tightly
Shifting fault	Connection piece worn	Replace
	Driving bevel gear loosen	Tighten the locking nut
	Shifting arm hole worn	Replace
	Spring inside main shaft fail	Replace
	Main shaft loosen	Tighten the bolt of gearbox cover
	Shifting lever deformation	Correct shifting lever or replace
Reverse gear fault	Reverse gear fork worn	Replace the fork or adjust cable
	Reverse gear cable fail	Adjust cable or replace
	Reverse gear shaft loosen	Tighten the bolts of the shaft
	Reverse gear fork jam	Clear the fork and connected parts
Reverse gear cannot rebound	Reverse gear shaft loosen	Tighten the bolts of the shaft
	Spring inside reverse shaft fail	Replace
	Reverse gear shaft deformation	Replace
Abnormal noise of gears	Bevel gear shaft, reverse gear shaft deformation	Replace
	Gear worn, too much clearance	Replace
	Bevel gear shaft, reverse gear shaft loosen	Replace
Main shaft and cover leakage	O-ring of main shaft fail	Replace
	Oil seal of main shaft fail	Replace
	O-ring of cover fail	Replace
Reverse shaft leakage	Reverse shaft bolt loosen	Tighten the bolt
	O-ring of reverse shaft fail	Replace

Description	Reason	Solution
Reverse fork leakage	O-ring fail	Replace
Clutch fork leakage	O-ring fail	Replace
Clutch shaft leakage	O-ring fail	Replace
Flange plate leakage	Bolt loosen	Tighten the bolt
	Gasket broke	Replace
Gearbox leakage	There is hidden void	Weld or paint to repair

4. Output system malfunction and solutions.

Description	Reason	Solution
Abnormal sound of gears	Gear worn or not assembled in order	Assemble again or replace gear
Gear running resisted	Assembled not in order	Assemble again
Overheat	Lacking of oil in gearbox	Fill the oil to the level mark
	Narrow gear clearance	Assemble again
	Narrow axial direction clearance	Assemble again
Leakage between gearboxes	Connecting bolts loosen	Tighten the bolts
	Gasket broke	Replace
PTO outer ring leakage	Oil seal fail	Replace
PTO inner hole leakage	Shaft sheath broke	Replace

5. Other malfunction and solutions.

Description	Reason	Solution
Blades broke	Hit rocks or other hard objects	Replace
Cable snap	Worn by time	Replace

Mini Tiller Warranty Period and Warranty Policy Guide Lines

Warranty Period

1. Mini Tiller Machine (Engine, Transmission) 06 months from date of invoice to buyer.
2. Normal maintenance and adjustments to machine is the responsibility of the customer.
3. Normal wear and tear is not covered under warranty.

No Warranty on Electrical Motor / Electrical Parts / Battery Etc.

Warranty Policy Guide Lines:

1. The Warranty period shall start from the date of purchase of the product and applicable to the first purchaser on receipt of the warranty Registration Card.
2. The warranty only covers the manufacturing defect or defective part within the specified warranty timeframe only.
3. The warranty will cease if Service maintenance is not carried out as per periodical maintenance schedule.
4. Warranty voids if Farmworld genuine parts or approved blades are not used on machine during periodical maintenance.
5. The warranty will not cover for normal wear & tear of parts such as brake, clutch plate, Spark-plugs, nylon line, air-filters, fuel filters, oil seals, drive belt, rubber parts, plastic parts, electrical parts.
6. Warranty for Proprietary parts, Like: Fuel Injection Pump, Fuel Injector, Tyre, Battery, Alternator & Starter Motor etc. are subject to respective manufacturers as per their warranty policy.
7. Under warranty period, only repair or replacement of defective parts will be considered, replacement of machine is not warranted.
8. Farmworld or its manufacturers will not be liable for any general damages, including injuries to body, or incidental or consequential damages or other commercial loss or damages.
9. Any Implied warranties, obligations or liabilities, including but not limited to any implied warranty of merchantability shall be limited in duration to the applicable warranty period.
10. Warranty is not transferable to the second and subsequent owners. No agent, representative, dealer or employee of its manufacturers has the authority to increase or alter this warranty.
11. The warranty ceases if the machine is serviced or repaired by unauthorized persons or at any Local workshop.
12. The warranty will not cover for any accidental damage, fire, collision, improper usage, tampering, customer abuse and external modifications/fabrications done by the customer.

Receipt is required for availing warranty services

Jurisdiction: All disputes are subject to Rajkot court's jurisdiction.

FARMWORLD reserves the right to refuse warranty if the product was found not to be carried periodical maintenance & installed as per out installation instructions.

What is covered:

Farmworld Agrotech Pvt. Ltd. and its manufacturers warrant this product to be free from defects in material or workmanship.

- (1) Parts defective in material and workmanship are covered.
- (2) Warranty will only cover defects arise under normal usage.
- (3) Warranty is limited to repair or replacement by Farmworld Agrotech Pvt. Ltd. or its manufacturers at the premises of Authorized Dealers, of such parts as appear to Farmworld or its manufacturers, upon inspection, to be defective in material and/or workmanship.
- (4) Farmworld or its manufacturers make no warranty with respect to trade accessories not manufactured or sold by them.

What is not covered:

The *warranty shall become null and void* and neither Farmworld Agrotech Pvt. Ltd. nor any of its manufacturers, nor its authorized dealers assumes any responsibility, if the failure was caused by the following:

- (1) Operation of product with incorrect fuel or lubricants.
- (2) Incorrect usage of machine or misuse.
- (3) Lack of maintenance.
- (4) Negligence.
- (5) Accident or physical damage.
- (6) Repairs made by unauthorized parties and/or with unauthorized parts.
- (7) Improper set up, adjustments, tampering or altered products.

Note: *The purchase is not dependent upon a product demonstration. The purchaser shall satisfy himself with the product, including any product demonstration or verification of any function, before buying. Farmworld Agrotech Pvt. Ltd. or its authorized dealers, including online sellers, shall not be liable to give any on-site demonstration after purchase of any of the product.*



Honour Your Trust

WARRANTY REGISTRATION CARD

(Company Copy)

Mini Tiller Details			
Model:		Invoice No.:	Invoice Date:
Engine Head No.:			
Engine Chassis No.:		Warranty Start Date:	Warranty End Date:
Please mark ✓ the machine is using for. List of accessories /Attachments	Ridger ☐	Plough ☐	Seed Drill ☐
	Paddy Wheel ☐	Weeding Blade ☐	Potato Digger ☐
	Trailer ☐	Water Pump ☐	
The Mini Tiller & accessories have been handed over to the Customer in good condition after pre-delivery inspection, training & service.			
Use of Mini Tiller	Farm Use (Rotavation)		
	Pudding	Sowing	Pumping Water
	Inter Culture	Harvesting	Spraying
	Other Use		
	Stone Cutting	Haulage	
Customer Details			
Name:			
Village:		PIN:	
City:		District:	
State:		Country:	
Contact No.:		Date of Birth:	
Email ID:			
Customer Signature (I acknowledge that I have read, understood and I accept the warranty policy and terms.)		Dealer's Seal & Signature	

*Dealer has to maintain the photo copy of the Warranty Registration Card.



Honour Your Trust



Honour Your Trust

WARRANTY REGISTRATION CARD

(Dealer Copy)

Mini Tiller Details			
Model:		Invoice No.:	Invoice Date:
Engine Head No.:			
Engine Chassis No.:		Warranty Start Date:	Warranty End Date:
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The Mini Tiller & accessories have been handed over to the Customer in good condition after pre-delivery inspection, training & service.			
Use of Mini Tiller	Farm Use (Rotavation)		
	Pudding	Sowing	Pumping Water
	Inter Culture	Harvesting	Spraying
	Other Use		
	Stone Cutting	Haulage	
Customer Details			
Name:			
Village:		PIN:	
City:		District:	
State:		Country:	
Contact No.:		Date of Birth:	
Email ID:			
Customer Signature (I acknowledge that I have read, understood and I accept the warranty policy and terms.)		Dealer's Seal & Signature	

*Dealer has to maintain the photo copy of the Warranty Registration Card.



Honour Your Trust

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	Other Use		
	Stone Cutting	Haulage	
Customer Details			
Name:			
Village:		PIN:	
City:		District:	
State:		Country:	
Contact No.:		Date of Birth:	
Email ID:			
Customer Signature (I acknowledge that I have read, understood and I accept the warranty policy and terms.)		Dealer's Seal & Signature	

*Dealer has to maintain the photo copy of the Warranty Registration Card.



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FARMWORLD AGROTECH PRIVATE LIMITED

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Taluka - Lodhika, District - Rajkot, 360021 (Gujarat) India.

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